Two-robot pick and place of a U-profile hash assembly

Summary: the robots only drop each profile into the open clamps of a pallet fixture (about 8 mm of play); when the clamps close, the fixture locates it to the 0.05 mm requirements; pre-machined parts supply the form features (height, flatness, straightness) that neither robot nor fixture can deliver.

- **Cell:** a two-station conveyor line. Station 1: R1 (xArm6) places both Pos 1. Station 2: R2 places both Pos 2. A simple conveyor stop holds each pallet; no precise location is needed. The clamps are spring-closed, so each pallet leaves the cell clamped for joining elsewhere. Estimated cycle time about 18 s per assembly.
- **Fixture:** passive fixed stops and spring plungers align one side of each part; actuated clamps align the other. A one-sided clamp on the lower Pos 1 is coupled 2:1 to a centring pair on the upper Pos 1, so the 120 mm spacing holds whatever the profile width.
- **Parts:** cold-formed profiles miss the 0.05 mm requirements by factors of 2 to 24. They are straightened and machined (height, contact spots, one end) before they reach the cell.

1 Part tolerances and assumptions

The cross-sections follow EN 10162 and the lengths ISO 2768-m. This allows ± 0.75 mm on web width, ± 1.2 mm on flange height, $\pm 2°$ on the bend angle, a bow of $0.0025 \times$ L (0.8 mm on Pos 1), a twist of 1 ° per metre and ± 0.5 mm on length, i.e. 2 to 24 times the 0.05 mm requirements of the assembly.

Assumptions

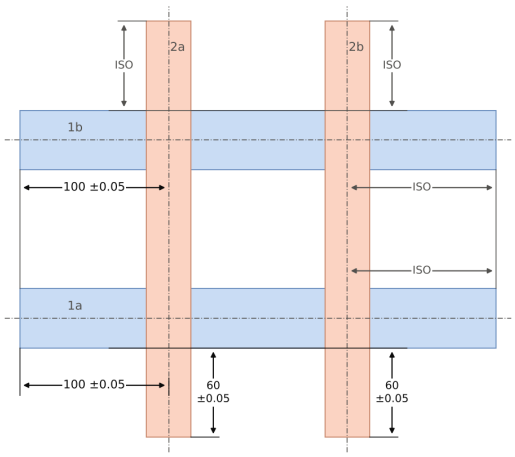


Fig. 1 How I read 2×100 and 2×60: black = ± 0.05 from the left / lower ends, grey = opposite ends, set by the part length (ISO 2768-m).

- **2×100 and 2×60** are read as in Fig. 1. This should be confirmed with the designer, who knows the intent of the assembly.
- **No joining in this cell.** The task asks for two pick-and-place robots and the drawing does not name a joining method. The pallet keeps all parts clamped and goes to a separate joining station.
- **Twist does not affect parallelism or perpendicularity.** Twist rotates the section by at most 0.32 ° (1 °/m over 320 mm). Tipping about one flange tip, the other tip, 40 mm away, shifts sideways by only $40 \times (1 - \cos 0.32°) \approx 0.0006$ mm, so the flanges stay where the fixture holds them and the two Pos 1 stay parallel. What twist does change is height: that tip lifts by $40 \times \sin 0.32° \approx 0.22$ mm, which the pre-machining removes (section 3).

2 Pick-and-place cell and fixture

- **Layout.** The two xArm6 face each other across a conveyor with two stations (Fig. 2). R1 serves station 1 (Pos 1), R2 serves station 2 (Pos 2), so the robots never share a workspace. Each robot picks from a gravity chute; the chute does not have to be very precise (about ± 2 mm), because the open fixture clamps have play. Payload (0.38 kg max.) and reach (700 mm) are no constraint.
- **Pallet location.** A retractable stop halts the pallet and holds it while the robot works; side guides keep it square on the belt (about ± 1 mm). Each robot is taught one station frame once.

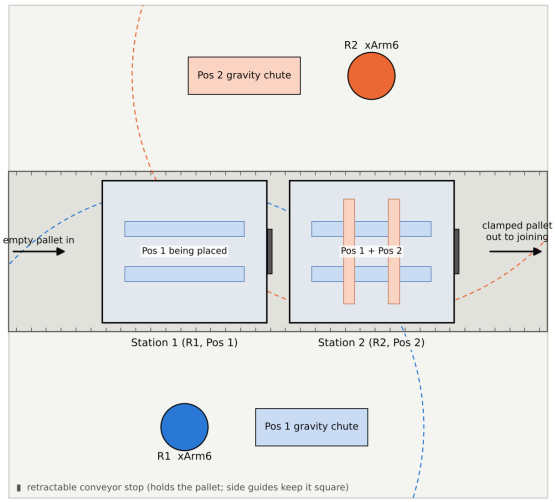


Fig. 2 Line layout, top view (mm, schematic). Dashed circles: usable reach at table height.

- **Tool.** Two flat suction cups on the web, which fit both profiles and need no clearance around the fixture jaws.

Fixture

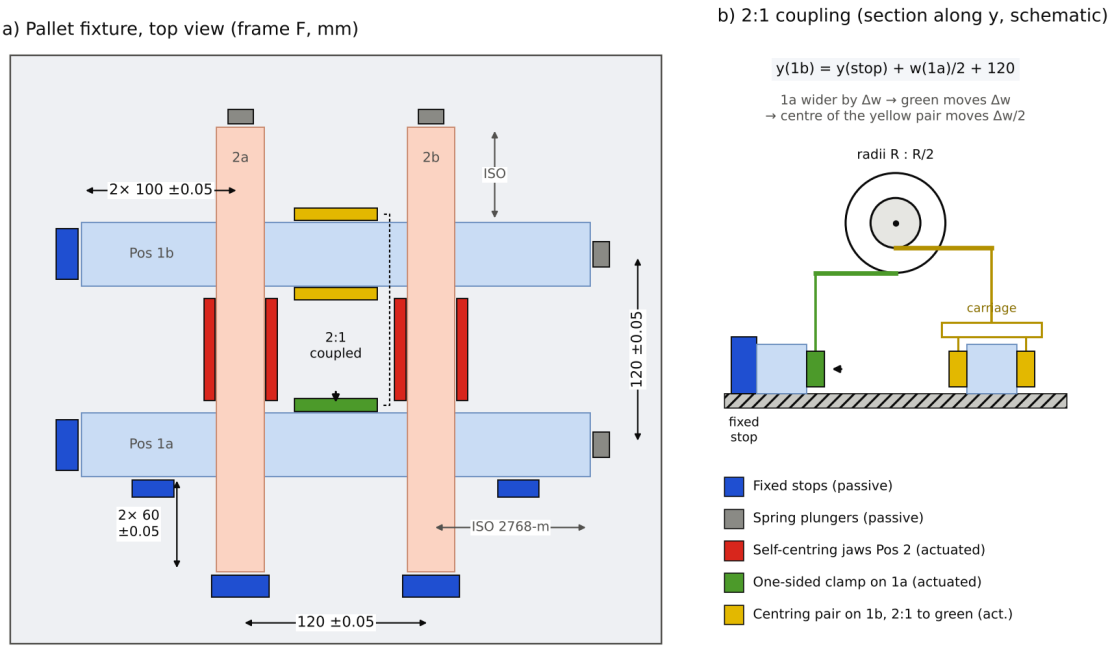


Fig. 3 Pallet fixture. a) Top view: the toleranced dimensions run from fixed stops, the free ends are ISO toleranced. b) Coupling between the green clamp and the yellow centring pair.

- **Passive vs. actuated.** Fixed stops and spring plungers are passive. The red jaws, the green clamp and the yellow pair are actuated. Open, they leave about 8 mm of play, so the robot only has to drop the part inside.
- **Pos 1a** is pushed by the green clamp against two fixed stops on its outer edge, the reference for both 60 mm dimensions.
- **Pos 1b** is centred by the yellow pair. The centre of this pair sits on a carriage driven 2:1 by the green clamp: if 1a is Δw wider, the green clamp moves by Δw and the centre of 1b by $\Delta w/2$. The spacing between the centrelines therefore stays at 120 mm for any width within the ± 0.75 mm tolerance.
- **Pos 2** is centred by self-centring red jaws with fixed centres 120 mm apart, which run the full length of the gap between the two Pos 1 so that they also align the angle.
- **Ends:** fixed end stops set the 2×100 and 2×60 dimensions; passive spring plungers at the free ends push the parts onto them.

Sequence

- **Station 1:** the station opens the Pos 1 clamps. R1 lowers 1a. The free end meets the chamfered spring plunger, which is pushed back as the part goes down. When the vacuum releases, the spring slides the part onto its end stop: **the first side is aligned**. R1 does the same for 1b. Then the green clamp and the coupled yellow pair close: **the other side is aligned**. The pallet moves on.
- **Station 2:** the same for 2a and 2b, with the red jaws closing at the end.
- **Cycle time.** Each station needs about 2×6.5 s for its two parts plus about 1.5 s of clamping. Adding the conveyor move gives about 18 s per assembly, with both stations working in parallel (estimate at 250 mm/s travel speed).

Software

- **Structure.** One program per station, written in Python with the UFACTORY xArm SDK. A station waits until its pallet is at the conveyor stop, places its two parts, closes the clamps and reports done; a small PLC-style handshake moves the conveyor once both stations are done.
- **Place move per part.** Pick from the chute by vacuum, move above the station, lower the part shifted slightly towards the spring plunger, release, retract. The fixture does the rest, so no vision or force control is needed.
- **Parametric.** Part sizes and place poses are read from a parameter file, so other profile sizes need new parameters rather than new code. Each station frame is taught once from three reference points.
- **Before the real cell:** a dry run of the whole sequence without hardware, checking every target against the robot's reach.

3 Part pre-processing (before the cell)

The fixture (section 2) delivers the positions and angles. Two requirements depend on the part geometry itself and are still open: the **four contact gaps $\leq$ 0.05 mm** and the **top surface of Pos 2 parallel to the base plate within 0.05 mm**. With raw profiles (section 1) neither is reachable, so the parts are prepared before they reach the cell:

- **Straighten** each profile in a press to about 0.02 mm of bow and correct the bend angle.
- **Mill the flange tips to height** (20.00 / 15.00 mm, $\pm$0.01), so every part has the same height and flat tips. The stock is ordered about 1 mm taller (e.g. flange 21 mm), because material can only be removed. This gives the top surface parallel to the base plate and removes the height difference caused by twist.
- **Skim the contact spots** on the Pos 1 web, where Pos 2 sits (a few hundredths of a mm), so flat flange tips meet a flat spot and the gap closes. Needed only if measured webs are not flat enough already.
- **Face the referenced end square**.

4 Worst-case check

Assumptions behind the table (worst case: all errors add up in the same direction):

- **Fixture:** stops and jaw centres machined and measured to $\pm$0.01 mm; jaw orientation held to 0.01 mm over the full part length; each centring jaw pair repeats to $\pm$0.005 mm; base plate flat to 0.005 mm.
- **Parts after pre-processing:** straightness 0.02 mm; flange tips at height $\pm$0.01 mm and flat to 0.01 mm; contact spots on the Pos 1 web flat to 0.01 mm; Pos 2 web top flat and untwisted; referenced ends square to 0.005 mm.
- **Robot side:** repeatability $\pm$0.1 mm (datasheet), absolute error after station teaching $\pm$0.2 mm, pallet at the stop $\pm$1 mm, part in the chute $\pm$2 mm, slip on the suction cups $\pm$0.1 mm. Along the length the part is set down further from the end stop than this total.

Requirement	Worst-case contributions	Result
120 $\pm$0.05, Pos 1	stops 0.01, carriage 0.01, centring 0.005	$\pm$0.025
120 $\pm$0.05, Pos 2	$2 \times$ (jaw centre 0.01 + centring 0.005)	$\pm$0.03
2×100 $\pm$0.05	jaw centre 0.01, centring 0.005	$\pm$0.015
2×60 $\pm$0.05	stops 0.01	$\pm$0.01
Pos 1 // Pos 1, 0.05	$2 \times$ 0.01 (stops) + 0.01 (yellow pair)	0.03
Pos 2 $\perp$ Pos 1, 0.05	$2 \times$ 0.01 (stops) + 0.01 (red pair)	0.03
Gap $\leq$ 0.05 (4$\times$)	flatness 0.01 + 0.01	$\leq$ 0.02
Placement (robot side)	robot 0.1 + absolute 0.2 + pallet stop 1 + chute 2 + cup slip 0.1 = $\pm$3.4, inside the 8 mm play of the open clamps	$\pm$3.4 (fits)

5 Validation on the real cell

- **Capture range:** offset the place pose in x, y and yaw on purpose and find where insertion fails. The target is reliable alignment at three times the expected placement error.
- **Fixture repeatability:** clamp the same set of parts 30 times, measure each time on a CMM, and check that the scatter and the offset from nominal are small compared with ± 0.05. Repeat with parts of different widths to check the 2:1 coupling.
- **Pre-processing:** measure a batch before and after straightening and machining to confirm the values assumed in section 4. Measure again after joining, to check that joining keeps them.

References

- UFACTORY xArm 6: 5 kg payload, 700 mm reach, ± 0.1 mm repeatability (ufactory.cc; docs.xarm.ufactory.cc).
- EN 10162:2003, Cold rolled steel sections: technical delivery conditions, dimensional and cross-sectional tolerances. DIN 59413, its national predecessor.
- ISO 2768-1, general tolerances (class m: ± 0.5 mm for 120–400 mm). ISO 1101, geometrical tolerancing.